

## Class 24: Bases of a Vector Space

**Definition:** If  $H$  is any subspace of a vector space  $V$ , a **basis** for  $H$  is a set of vectors in  $H$  that

Recall that a set  $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p\}$  is linearly independent if

**Practice:** Show that the set of functions  $\{1, \cos(t), \cos^2(t)\}$  is linearly independent.

## Standard basis sets for common vector spaces

For  $\mathbb{R}^n$

For  $\mathbb{P}^n$

For  $M_{2 \times 2}$

### Practice:

1. Find a basis for the space spanned by the vectors given below.

$$\begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} -2 \\ 0 \\ 0 \\ 2 \end{bmatrix}, \begin{bmatrix} 3 \\ -1 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 5 \\ -3 \\ 3 \\ -4 \end{bmatrix}, \begin{bmatrix} 2 \\ -1 \\ 1 \\ 0 \end{bmatrix}$$

2. Find a basis for  $\mathbf{R}^4$  that includes the vectors

$$\begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix} \text{ and } \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \end{bmatrix}$$

3. Consider the polynomials  $\mathbf{p}_1(t) = 1+t$ ,  $\mathbf{p}_2(t) = 1-t$ , and  $\mathbf{p}_3(t) = 2$ . Explain why these three vectors cannot be a basis for  $\mathbf{P}^1$ . Show that these three vectors are linearly dependent.

**Proof Practice:** Suppose  $T$  is a one-to-one linear transformation. This means that if  $T(\mathbf{u}) = T(\mathbf{v})$  we know that  $\mathbf{u}$  must equal  $\mathbf{v}$ . Prove that if the set of images  $\{T(\mathbf{v}_1), \dots, T(\mathbf{v}_p)\}$  is linearly dependent, then  $\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$  must be linearly dependent.

Proof: